

Ayush Hrishikesh Mishra

Bengaluru | h.ayushm@gmail.com | linkedin.com/in/ayush-h-mishra | Portfolio | 6376268571

Technical Skills

Languages: Python, Java, SQL

Web Frameworks: Next.js, React.js, Flask, Fast-API

AI/ML & Agents: Machine Learning, Deep Learning, Generative AI, RAG, Lang Graph, Crew.ai

Databases: MongoDB, MySQL, ChromaDB

Cloud & Tools: Docker, Git, REST-API, Graph-ql, HuggingFace

Experience

Software Engineer,

Jul 2024 – Present

Volkswagen Group Digital Solutions, Bangalore

- Implemented an end-to-end assistant to automate project scoping for **10+** business units, generating comprehensive PRDs from user requirements.
- Migrated multi-agent orchestration from Langflow to LangGraph, implementing PostgreSQL for persistent inter-session memory and state management.
- Optimized ingestion pipelines by parallelizing document extraction tasks, reducing total processing time by **62%**.
- Augmented Crew.ai workflow by integrating PostgreSQL memory to generate PRDs persisted to **AWS S3**, reducing manual drafting time by **90%**.
- Engineered asynchronous task distribution using AWS SQS and DLQs to ensure 100% reliability for over 50+ daily message transactions.
- Developed a Graph-RAG platform using LLM pipelines and Neo4J, cutting project discovery time by **60%**.
- Integrated real-time Knowledge Graph visualization, allowing stakeholders to trace **10+** relationship types and accelerating decision-making by **25%**.
- Enhanced AI response trust with citations and similarity scoring, boosting user adoption by **40%**.
- Orchestrated a multi-agent system using Crew.ai and an LLM-based Manager Agent to automate IT workflows, resulting in a **35%** reduction in manual coordination.

Software Engineer Trainee,

Dec 2023 – Jun 2024

Volkswagen Group Digital Solutions, Pune

- Built a RAG chatbot serving **5,000+** users with context-aware responses and citations, enabling faster and more accurate knowledge retrieval.
- Processed **200+** PDFs via AWS Textract and generated embeddings with BGE stored in ChromaDB.
- Used MySQL to capture user feedback, driving continuous improvements and iterative model refinement.
- Designed an interactive chatbot user interface showing conversations with citations, increasing user engagement by **90%** and adoption across teams.
- Awarded **Innovation Award** for transforming enterprise knowledge access and demonstrating measurable impact on internal workflows.

Education

B.Tech Computer Science and Engineering with Specialization in Robotics

2020 – 2024

Vellore Institute of Technology-AP

Amaravati, Andhra Pradesh

Publication

Cognitive Assistance for Dementia Patients (IEEE)

Mar 19, 2024

2024 ASU International Conference on Emerging Technologies for Sustainability and Intelligent Systems (ICETSIS), Manama, Bahrain.

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